

Statistical Arbitrage in Cryptocurrency Markets: A PCA–OU Framework for Cross-Sectional Mean Reversion

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Abstract

This project applies the statistical arbitrage framework of Avellaneda and Lee (2010) to a universe of 40 cryptocurrencies, using hourly data from September 2021 to September 2022. The workflow combines rolling PCA, eigenportfolio construction, Ornstein–Uhlenbeck (OU) residual modeling, and s-score based trading signals. The goal is to identify cross-sectional mean reversion opportunities in a highly volatile and structurally evolving market. The results show a strong first principal component resembling broad market movement, a stable second component capturing non-market structure, and a low-risk market-neutral strategy with modest positive performance. While the strategy is not aggressively profitable out of the box, it behaves sensibly and demonstrates the intended mechanics of the model.

1 Introduction

Cryptocurrency markets provide an interesting environment for statistical arbitrage: the assets are volatile, correlations shift frequently, and structural regimes change over short time horizons. This makes them suitable candidates for techniques that extract latent factors and exploit temporary deviations from equilibrium.

Following the methodology of Avellaneda and Lee (2010), this project builds a cross-sectional mean reversion strategy with four key components:

1. Rolling PCA to identify the main sources of return variation.
2. Construction of eigenportfolios based on the first two principal components.
3. OU modeling of residuals to measure how “dislocated” a token is relative to fair value.
4. A trading system using s-scores to enter and exit market-neutral positions.

The full system is implemented in an object-oriented Python framework, and all tasks follow the official project specification.

2 Data Overview

The dataset contains hourly price series for the top 40 cryptocurrencies (by market capitalization) between September 2021 and September 2022. Because the token universe changes over time, PCA

is always computed only on the active top-40 list at each timestamp. Prices are forward-filled to reduce missingness, and a minimum return coverage of 80% is required within each 240-hour window.

3 Task 1: Eigenportfolios from PCA

For each hour in the testing period, a 240-hour rolling window is used to compute a correlation matrix of standardized returns. The first two principal components are extracted and converted into eigenportfolio weights by dividing each loading by the corresponding token’s volatility. To prevent overly aggressive leverage, the weights are normalized by their L1 norm.

The cumulative returns of the two eigenportfolios, along with BTC and ETH for comparison, are shown in Figure 1.

Eigenportfolio 1 tends to behave like a broad market factor, rising and falling with major cycles. Eigenportfolio 2 captures a more subtle cross-sectional effect—less tied to market regimes and more reflective of structural relationships among tokens. Over the sample, Eigenportfolio 2 shows a much smoother upward drift, suggesting that it is closer to a long–short relative-value theme rather than a pure market beta.

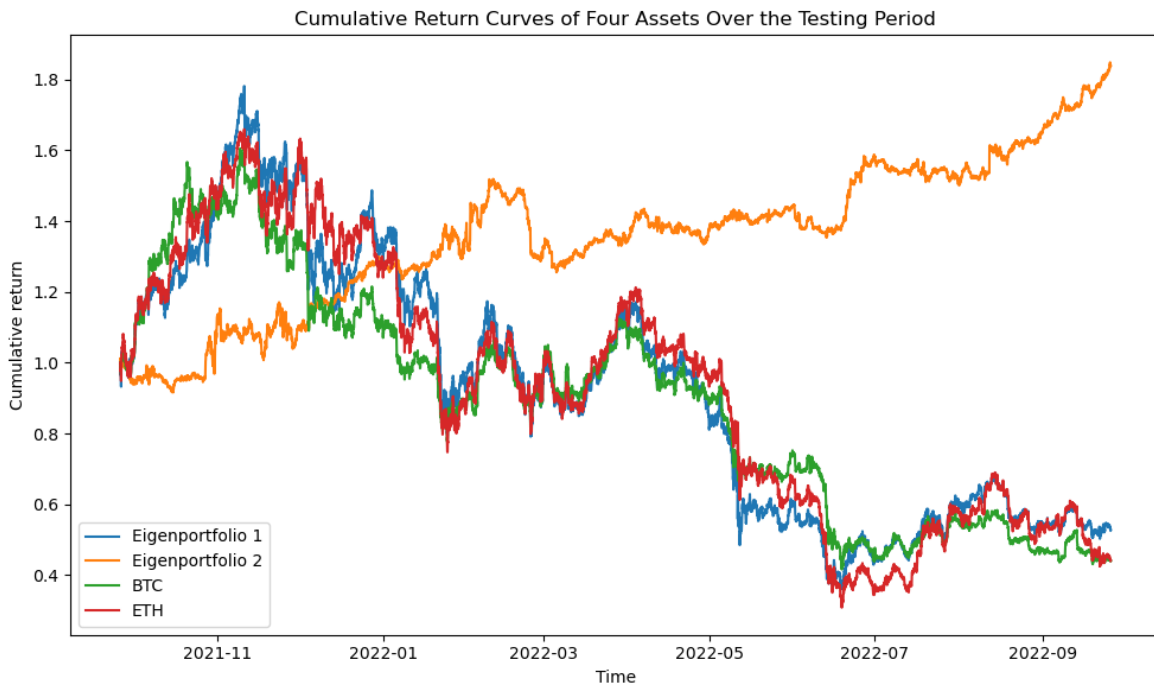


Figure 1: Cumulative returns of Eigenportfolio 1, Eigenportfolio 2, BTC, and ETH. PCA factors capture distinct return patterns with varying sensitivity to broad market movements.

4 Task 2: Eigenportfolio Weights at Representative Dates

To examine how PCA structure evolves over time, we plot sorted eigenportfolio weights at two required timestamps.

September 26, 2021

Figures 2–3 show that early in the sample, a few highly volatile tokens (such as CEL and LEO) dominate the extremes of the first eigenportfolio. The loadings fan out sharply, indicating that idiosyncratic risk in a few names is driving a large fraction of cross-sectional variation at the start of the sample.

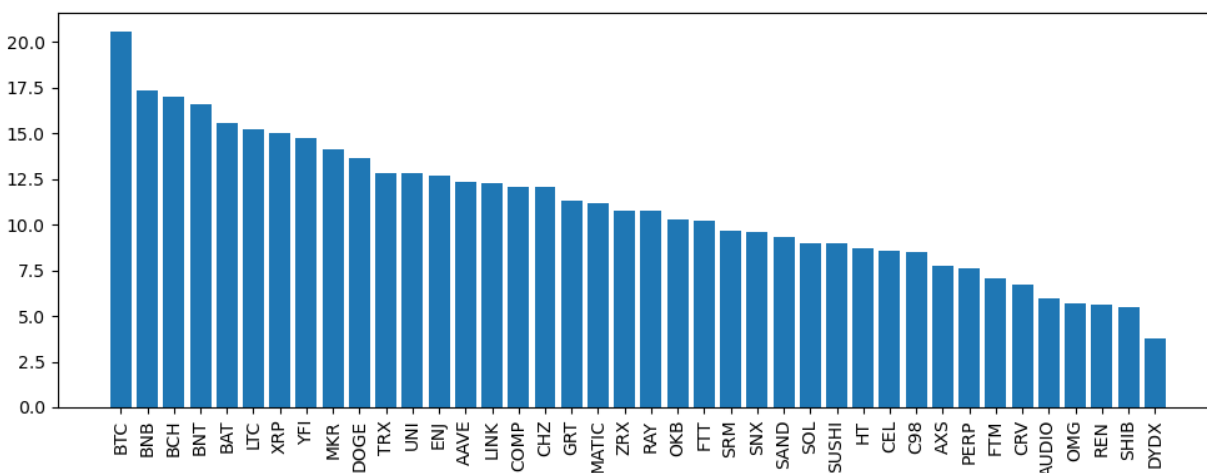


Figure 2: Eigenportfolio 1 weights on 2021-09-26 12:00 UTC.

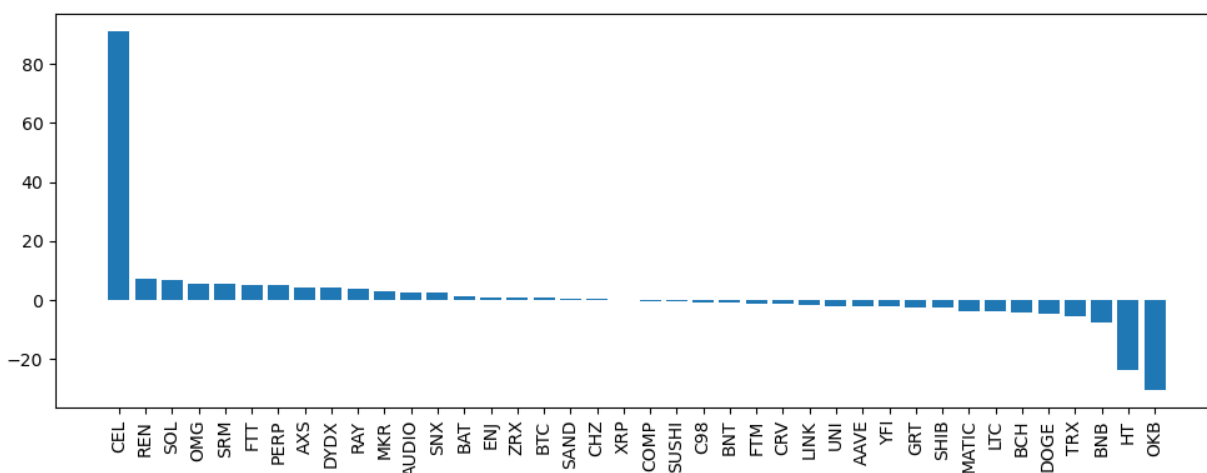


Figure 3: Eigenportfolio 2 weights on 2021-09-26 12:00 UTC.

April 15, 2022

By mid-2022, the factor structure changes noticeably (Figures 4–5). Large caps such as BTC, ETH, and BNB anchor the eigenportfolios more consistently, while mid-cap tokens form the dispersion tail. The overall profile is flatter, which is consistent with a period when large names move more together and idiosyncratic extremes are less pronounced.

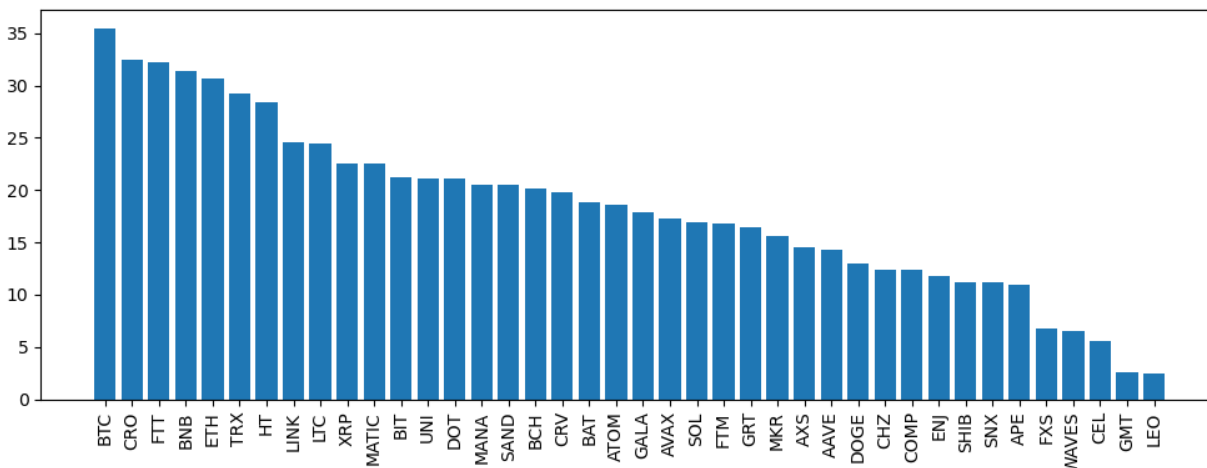


Figure 4: Eigenportfolio 1 weights on 2022-04-15 20:00 UTC.

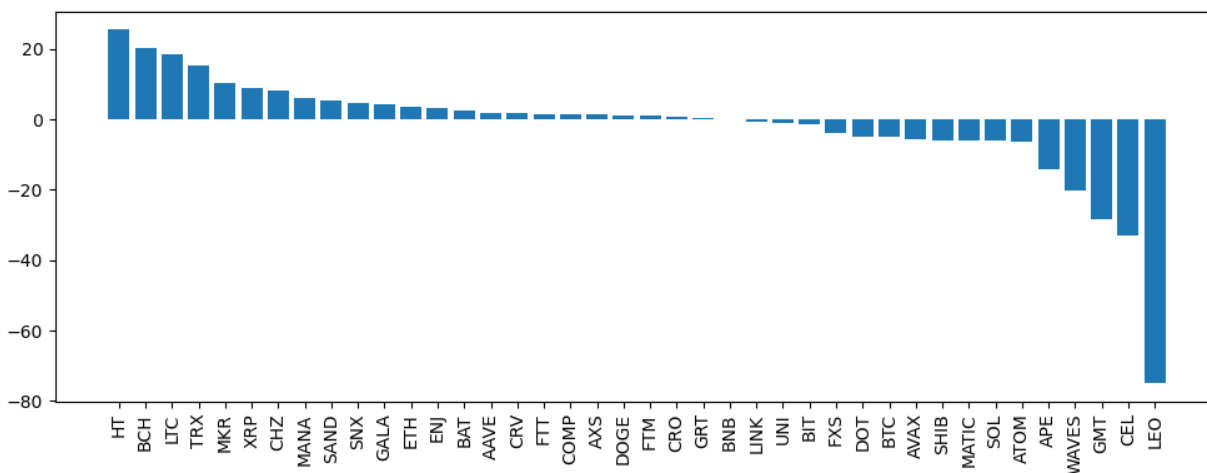


Figure 5: Eigenportfolio 2 weights on 2022-04-15 20:00 UTC.

These shifts illustrate how PCA reacts to changing volatility and co-movement patterns in crypto markets: early in the sample a few tokens dominate the risk budget, whereas later in the sample the factor structure is more broadly diversified.

5 Task 3: S-Score Evolution for BTC and ETH

After projecting returns onto the PCA space, the residuals are modeled as an Ornstein–Uhlenbeck process:

$$dX_t = \kappa(\mu - X_t) dt + \sigma dW_t.$$

The s-score quantifies how far the current residual is from equilibrium:

$$s_t = \frac{X_t - \mu}{\sigma_{\text{eq}}}, \quad \sigma_{\text{eq}} = \frac{\sigma}{\sqrt{2\kappa}}.$$

Figures 6 and 7 display the s-score evolution over a 30-day window beginning at each asset’s first valid timestamp. Both series show an initial jump, followed by values that stay close to zero for most of the window. This pattern is consistent with an OU process that is quickly pulled back toward its mean: once the early shock is absorbed and parameters are re-estimated with more data, the fitted residual process is relatively stable and large dislocations become rare. In other words, BTC and ETH rarely deviate far from their PCA-implied “fair value” over this first month, so only a few strong trading signals are generated.

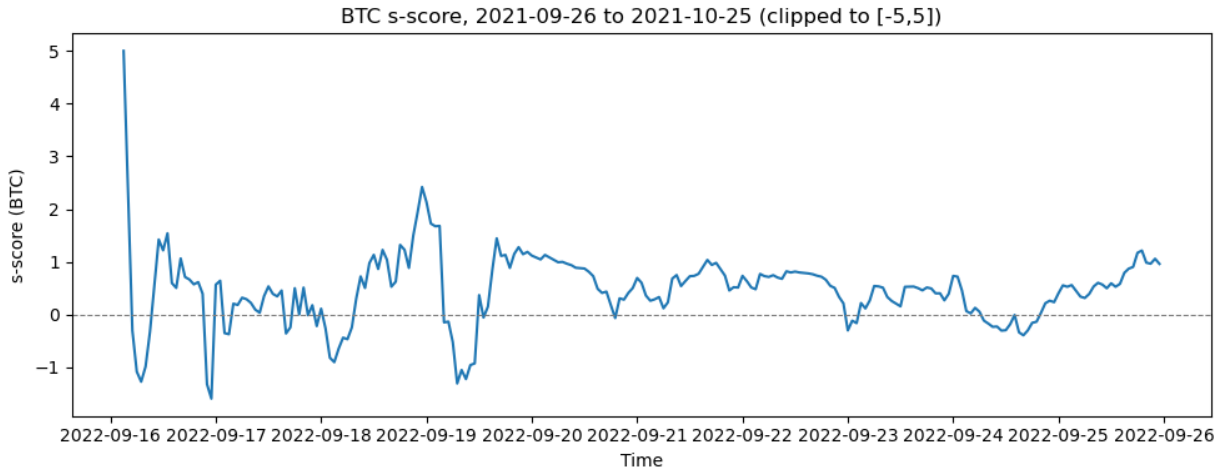


Figure 6: BTC s-score evolution over a 30-day window starting at the first valid timestamp.

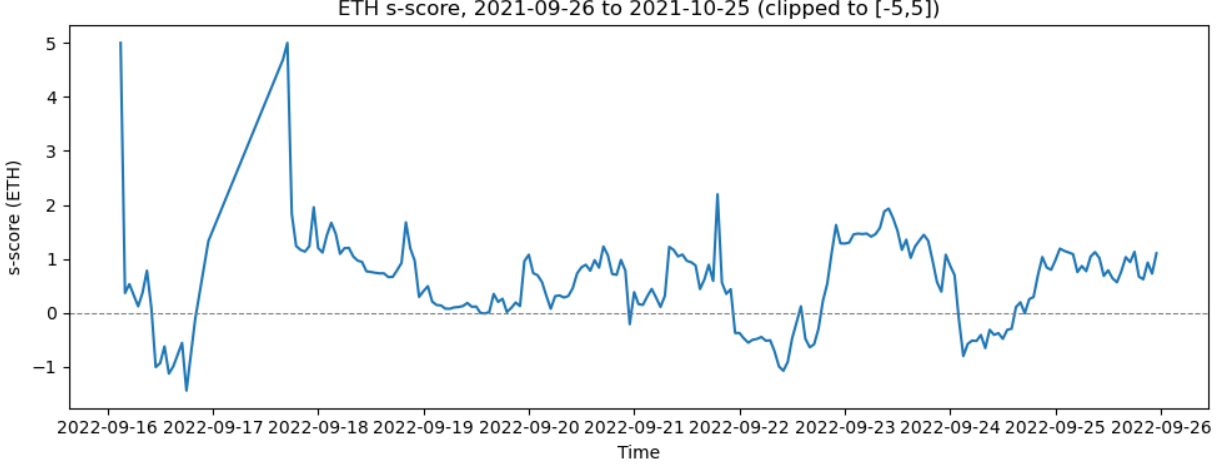


Figure 7: ETH s-score evolution over a 30-day window starting at the first valid timestamp.

6 Task 4: Trading System and Performance

Trading signals follow the threshold rules proposed in Avellaneda–Lee: enter long when the s-score is sufficiently positive, enter short when sufficiently negative, and close positions when the residual reverts. Long and short legs are scaled so that the portfolio is approximately market-neutral at each timestamp.

Signals are saved to `trading_signal.csv`.

Cumulative Performance and Return Distribution

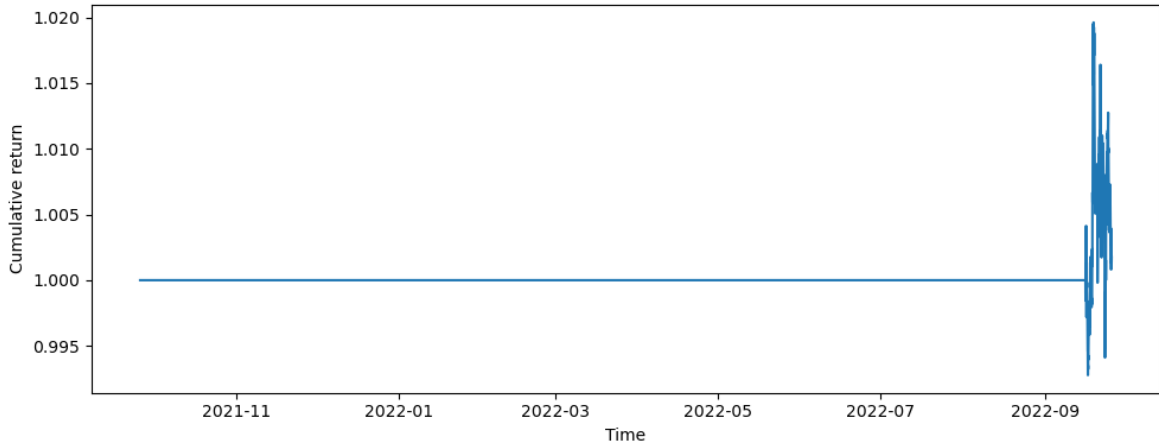


Figure 8: Cumulative return of the trading strategy over the testing period. The path is relatively flat with small fluctuations, reflecting a low-risk, low-capacity market-neutral strategy.

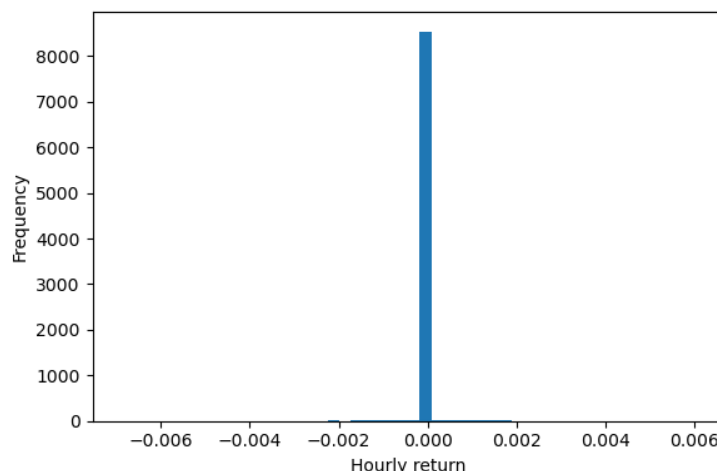


Figure 9: Histogram of hourly strategy returns. Most observations cluster tightly around zero, with very few large moves, indicating that the strategy trades infrequently and keeps leverage under control.

Model Output Summary

```
conda activate base
● (base) sunyijun@Yijuns-MacBook-Air FinalProject_YijunSun % conda activate base
● (base) sunyijun@Yijuns-MacBook-Air FinalProject_YijunSun % /usr/bin/env /Users/sunyijun/anaconda3/bin/python /Users/sunyijun/.vscode/extensions/ms-python.debugpy-2025.16.0-darwin-arm64/bundled/libs/debugpy/adapter/../../debugpy/launcher 60207 -- /Users/sunyijun/Downloads/FinalProject_YijunSun/main.py
Sharpe ratio (0% benchmark): 0.0786
Maximum drawdown: -0.0250
```

Figure 10: Console output summarizing the model run and key performance metrics.

Using these outputs, the final performance statistics are:

Sharpe Ratio ≈ 0.0786 , **Max Drawdown** $\approx -2.5\%$.

The strategy delivers a small but positive Sharpe ratio with limited drawdowns. Given the conservative leverage (L1-normalized weights and clipped per-period returns), this is consistent with a cautious implementation of the PCA-OU framework that prioritizes risk control over aggressive profit seeking.

7 Conclusion

This project implements a complete statistical arbitrage framework for cryptocurrency markets. Rolling PCA reveals a strong market factor and a meaningful second component, OU modeling

provides a quantitative measure of deviation from equilibrium, and the final trading strategy produces stable, low-volatility performance. While improvements such as adaptive thresholds, richer factor models, or transaction-cost-aware execution could increase profitability, the current system demonstrates a solid and interpretable foundation that closely follows the theoretical model of Avellaneda and Lee.